

QUEENSLAND UNIVERSITY OF TECHNOLOGY

Investigating Applications Within Triboelectric Nanogenerators

Thomas Cox, Mahesh Y. Chougale, Niroshan Manoharan, Deepak Dubal

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Abstract

With the world constantly in motion around us, there is a surplus amount of mechanical energy going to waste. This work aims to explain the triboelectric effect and investigate a possible way to harness this waste energy through the form of triboelectric nanogenerators. First, appropriate triboelectric materials were selected, resulting in the use of a PVA/GNP nanocomposite (tribopositive) and SiS (Tribonegative). These materials implemented into 3.5x3.5cm device provided adequate electrical output for the testings of applications such as creating pedometers and digital switches. With this understanding in simplistic applications, other works were investigated to understand future applications for these devices within medicine, computer science, and high-voltage energy generation and gas detection.

Keywords: triboelectric nanogenerators, energy harvesting, renewable energy, nanotechnology

1 Introduction

Introduced in 2012 [1], triboelectric nanogenerators (TENGs) are a multilayered device capable of converting mechanical energy into electrical impulses. Since their discovery, these devices have increased in popularity and gained significant attention as a potential cornerstone of renewable nanoenergy [2]. The rising demand for energy-efficient and self-sustaining systems has made these devices highly desirable in fields such as wearable electronics, medical devices, and passive power system energy generation [3]. This work aims to investigate the mechanisms behind these devices, possible applications, and understand applications that are already in place.

2 Mechanisms of TENGs

2.1 Triboelectric Effect

As the name suggests, the main force behind TENGs is the triboelectric effect. This effect, also referred to as contact electrification, has been well documented since its initial discovery 2600 years ago [4] and is described as the generation of a triboelectric charge that is created when two dielectrics rub together. However, in 1789 this definition was changed as Alessandro Volta noted the charge transfer was generated by contact rather than rubbing [5]. This charge is created due to the differential in electrical properties, such as electron affinity or conductivity. This facilitates the transfer of free electrons, creating a net positive and negative electromagnetic potential between these materials [6]. This exchange of electrons is partially denoted by Eqn. 1 which describes how varying electric fields and polarisation can contribute to the generation of electric currents through the relationship between the displacement current density J_d , permittivity of free space, ϵ_0 , the rate of change of electric field, $\frac{\delta E}{\delta t}$ and the rate of change of polarisation, $\frac{\delta P}{\delta t}$.

$$J_d = \epsilon_0 \frac{\delta E}{\delta t} + \frac{\delta P}{\delta t} \quad (1)$$

Other fundamental equations for this include:

$$C = \frac{\epsilon S}{\lambda} \quad (2)$$

which explains how the capacitance of the device is related to the material's dielectric constant, ϵ , the surface area S , and the charge penetration depth λ which is given by:

$$\lambda = \frac{k}{4\pi} \quad (3)$$

where k is the electrostatic constant.

$$\sigma = \frac{F_{applied}}{A} + F_{adhesive} \quad (4)$$

This equation for the surface charge density, σ , shows that its relationship with the force applied, $F_{applied}$ to the contact area, A , and the adhesive force applied to the device, $F_{adhesive}$.

These four equations largely denote the mechanisms behind triboelectric nanogenerators showing that changes in the physical properties of these devices, will create a direct change in electrical output.

2.2 A Brief History of Triboelectric Applications

Over the millennia, systems using this principle have emerged, including Otto von Guericke's friction machine (a primitive form of an electrostatic generator) in 1663, Alessandro Volta and Abraham Bennet's influence machine in 1787, and Robert J. Van de Graaf's Van de Graaf generator in 1929 [5]. These devices have led to many important inventions such as particle accelerators, x-ray generators, and several nuclear physics advancements [7].

2.3 Triboelectric Modes

There are four primary TENG modes: Vertical contact-separation, lateral-sliding, single-electrode, and freestanding triboelectric layer [8]. For the purposes of these experiments, the vertical contact-separation mode was used for its simplicity and efficiency for being used as a low voltage-low current power source.

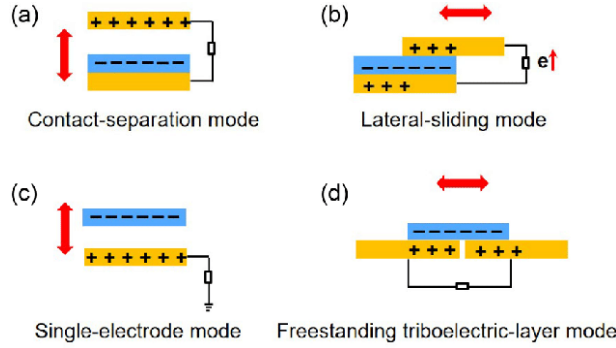


Figure 1: 4 Primary Modes of TENGs

2.4 Triboelectric Material Selection

Within triboelectrics, dielectric materials are displayed in triboelectric series (fig. 2) showing their affinity to obtain (tribopositive) or donate (tribonegative) electrons when in contact with other materials. For the purposes of this investigation, the materials used are displayed in fig. 3. These materials were selected due to their compatibility of conductivity, electron affinity, and triboelectric properties, with PVA/GNP being tribopositive and SiS being Tribonegative in relation to each other. Using these materials provided voltage and current readings that changed depending on things such as: material concentration, device size, and tapping frequency (fig. 4). For the purposes of our experimentation, a 3.5×3.5 cm devise with a 5% material concentration was used. This provided a compact device size that was capable of powering any low current portable device such as those discussed later on.

2.5 Synthesis of Materials

To create the PVA/GNP nanocomposite (fig. 5), PVA was added to a beaker filled 70% with water. GNP was then added and homogenised to create the nanocomposite that was capable of using the electrical properties of PVA that also benefits from the strong physical properties of GNP. This solution was then fed through an electron scattering process to evenly coat a sheet of aluminium foil.

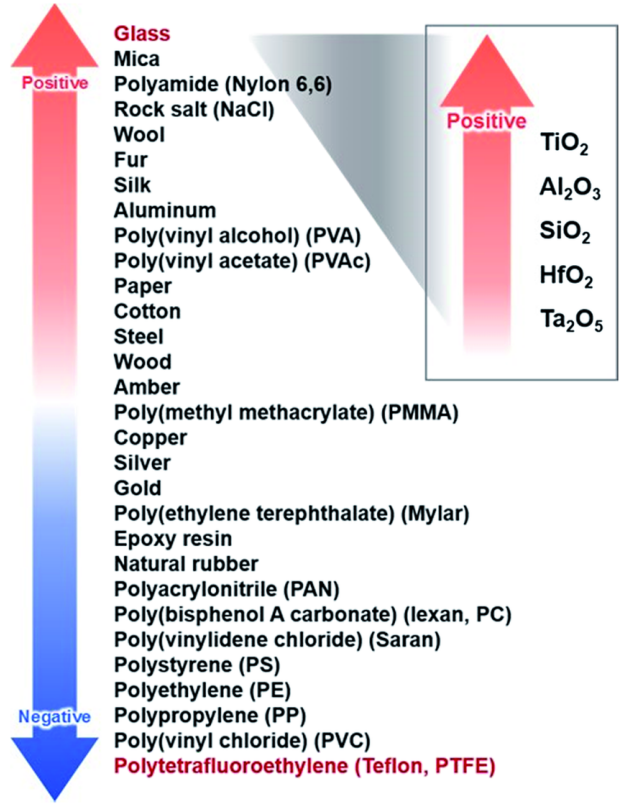


Figure 2: Example Triboelectric Series [9]

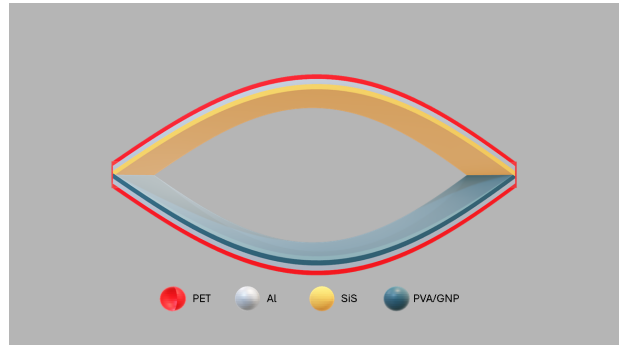


Figure 3: Structure of the Used TENG and its materials

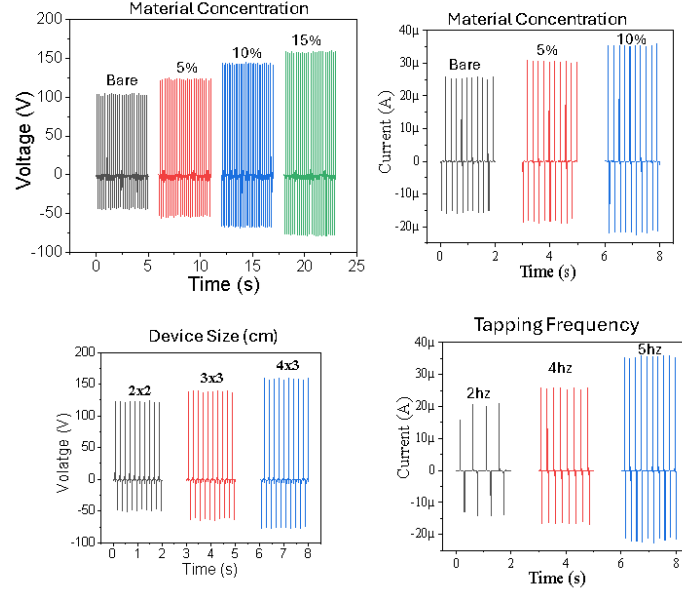


Figure 4: Voltage and current varying based on different physical properties

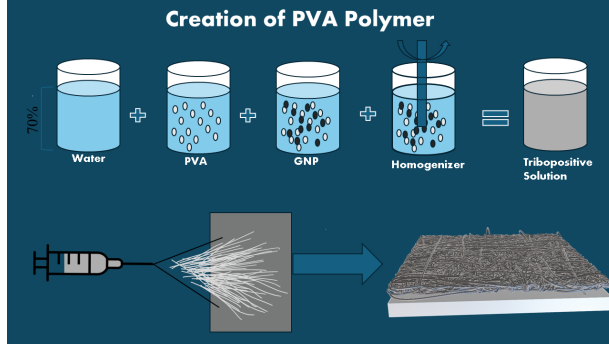


Figure 5: Creation of PVA/GNP Polymer Through Homogenisation and Electron Scattering

3 Application processes

The PVA/GNP, SIS, and Al materials were layered to form the full TENG structure (fig. 6a) [10]. After initial voltage readings, the device size was found to be suitable to be processed by an analogue port from an Arduino Nano (fig. 6b) once being fed through a load. In this case, a 10k ohm resistor in series with a small rectifier diode (1N4004) was used, as only positive voltages would be needed. Based on these results, two applications were investigated: the use of a TENG as a self-powering health tracker and the use of a new switch type to interface with elevators.

3.1 Self-Powered Health-Tracking Device Application

Whether medically recommended or simply for casual exercise, health-tracking devices have been very heavily popularised in recent years [11], with one example being the implementation of pedometers in many devices such as smartphones or watches. However, these devices, especially commercial ones, can suffer from inaccuracies [12]. So, with the idea to make a hyper-accurate pedometer that could, for example, fit in the sole of your shoe (fig. 6c), providing real-time data, the Arduino-connected TENG normalised to 0-5V was connected to MATLAB to pursue this idea. Initially, the program was made to handle a simple step counter using a while loop that, when detecting a voltage above 4.5V, would increment a counter titled “steps.” Although providing a very rudimentary process for a pedometer, many problems became apparent, such as incrementing multiple times for each step or not registering enough pressure to register above 4.5V; therefore, it did not count as a step. To fix this, the threshold was removed, and instead, a voltage peak approach was implemented. This program would increment the counter given two conditions: the voltage is above 2V, and the current reading is at least 1.5x greater than the previous. Implementing these conditions, along with other digital filtering for a cleaner signal, provided a much more responsive and efficient counter.

With this base counter implemented, further health tracking was desired through the categorisation of different “step types” to further track different exercises. The first two types, walking and running, were created alongside a figure that would display voltage readings and three counters: walking, running, and total steps (fig. 6d). Once created within the program, they were initially made to increment the counter depending on the voltage achieved per step, with lower voltages incrementing walking and higher voltages incrementing running. This, however, provided very inaccurate data, especially from person to person, as it would heavily depend on the force the person would put on their feet when running. Due to this oversight, the program was changed to increment the step type based on the frequency of steps,

with steps that have a frequency of 2Hz or less incrementing walking and anything greater than 2Hz incrementing running. This frequency threshold was implemented based on data on the average speed that people walk at [13]. Using this device provided data (fig. 6e) that suggests the success of a hyper-accurate pedometer, as each step would correctly provide an electrical pulse which in turn would increment the correct type of counter.

3.2 TENG as a Self-Powered Switch Type

Using the same TENG-Arduino device as before, testing of using it as a switch type began, with an initial application being that of an elevator switch simulated through the Unity 3D software interfacing with the Arduino IDE (fig. 7). This application program was more straightforward, as the previous filtering was used. Within Unity, conditions were established to make it possible for it to read the outputting serial monitor from the Arduino IDE, which was printing voltage values from analogue pins A1-A4, corresponding to floors G-3. Once a voltage was read above 2V, it would queue the floor position into an array that would cycle through these values. For example, if the switches were pressed that corresponded to floors G, 3, and 2 in that order, it would stop at the second floor before going to the third floor.

3.3 Future Improvements on Applications

3.3.1 Pedometer Device

The pedometer created, although capable of providing proof of concept, would be insufficient if it were to be used as theorised. Further improvements could include implementation of further step types such as: jumping, climbing, cycling, etc. Additionally, a counter could be added to assist in calorie counting based on commonly accepted values, such as those discussed by Hoi et al. [14]. To further cut down on size, this device could be connected to a PCB capable of processing all voltage data needed and relaying it to a Bluetooth device. These changes could make

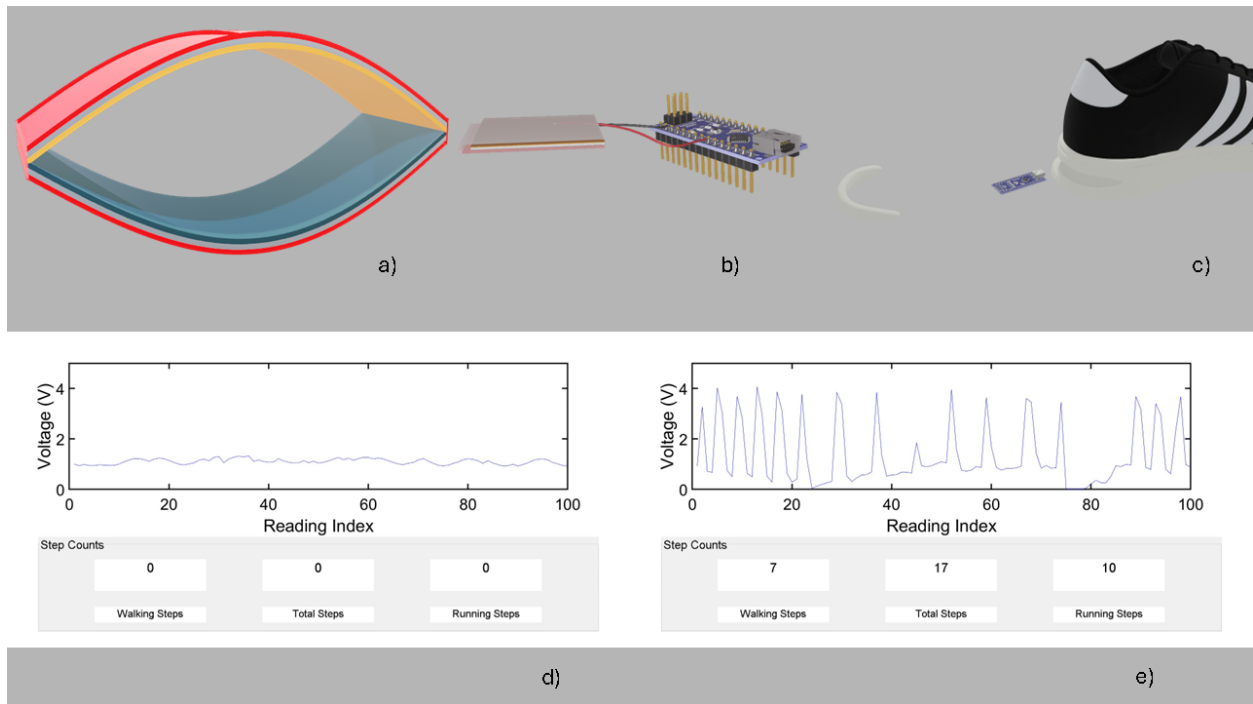


Figure 6: Health-tracking device prototype and results.



Figure 7: Elevator switch application using TENG technology.

for a suitable commercial product capable of accurate, real-time data for assistance in tracking health needs, whether it be a medical necessity or simply for recreational purposes.

3.3.2 Digital switch

The primary issue with the current digital switch device is the structure itself. Currently, the device has a malleable form capable of providing the bare necessities for a multitude of tests. If this device were to be used for real-world applications, it would need to be changed to better suit long term use. This could be done by replacing the malleable PET with a structure that is more rigid, such as PETG, to provide enhanced structural integrity (fig. 8). With this more secure structure, springs could be implemented to provide a stable position that is capable of being compressed to create the necessary voltage to activate the switch.

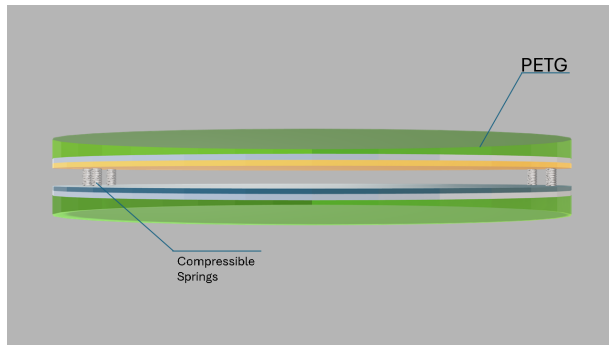


Figure 8: Example button with enhanced structural integrity

4 A Brief Investigation on More Complex Applications

4.1 Medical Applications

Some more advanced applications using more discrete mechanical energy have already been investigated.

Particularly within health sciences [15]. The applications that Yu et al. explore include applications in tissue regeneration and rehabilitation support.

4.1.1 Tissue regeneration

Within the body, as tissues regenerate, the areas around this regeneration are constantly in motion. This motion could be harnessed through the use of TENGs to stimulate regeneration. This is done by creating electrical stimuli that enhance the osteogenic process, which in turn facilitates bone growth. TENG devices have also been used in wound healing application where they create an electric field in the injured region that stimulate cell movement and growth factors which accelerate the healing process.

4.1.2 Rehabilitation

Similarly to the previous pedometer device, this application was investigated as a wearable TENG-based device that enable real time monitoring of physiological data. The device uses this data to create appropriate low frequency electrical stimuli which facilitate muscle and nerve growth, as is displayed by Scott et al. [16] which explains that chronic low frequency stimuli can increase resistance to fatigue, improve muscle efficiency, and improve muscle function.

4.2 High-Voltage Power source

Previous explorations of TENGs as a power source focused on lower power models using the vertical contact-separation mode. However, TENG applications for use as a high voltage power source have also been pursued [4]. By using a lateral-sliding mode TENG within an open circuit, voltages of several kilovolts were obtained, showing that TENGs can be used as a portable, cost-effective power source for high voltage systems such as HVAC systems.

4.3 Gas Detectors

TENG can also use their ability to act as a self-powered sensor to be utilised as gas detectors [17].

The design principle of these sensors is based on the change in electrical conductivity from the disturbance of ionised gas causing a significant reduction in voltage, allowing for measurements of concentration.

5 Conclusion

This work successfully investigated the mechanisms behind triboelectric nanogenerators as well as possible applications for them. These cost-effective devices were shown to be incredibly customisable to fit a number of purposes in fields such as medicine, computer science, and energy generation. Our PVA/GNP — SiS based device was able to provide sufficient voltage and current to the devices created. These devices relied purely on a voltage reading such that a low current, as is provided by our device, was capable of fulfilling all application requirements. Furthermore, recommendations for improvements on these devices have been provided with the idea of possible commercialisation and additional complex applications were discussed.

6 Author contribution statement

Thomas Cox: Conceptualisation, Formal Analysis, Methodology, Experiments, Writing — original draft. **Mahesh Y. Chougale:** Conceptualisation, Material Fabrication, Formal Analysis, Methodology, Experiments. **Niroshan Manoharan:** Conceptualisation, Material Fabrication. **Deepak Dubal:** Conceptualisation

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